Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is i^2? Student's answer: 1

Recall the definition of i.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
By definition $i^2 = -1$. The student dropped the negative sign.	**-1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify $\sqrt{(-9)}$. Student's answer: ±3

Write the square root of a negative number using i.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$\sqrt{(-9)} = \sqrt{9} \cdot \sqrt{(-1)} = 3i$. Real square roots of negatives do not exist.	**3i**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is i^3? Student's answer: i

Multiply by i each time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$i^3 = i^2 \cdot i = -1 \cdot i = -i$. The student forgot to apply $i^2 = -1$.	**$-i$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is i? Student's answer: i

Multiply by i each time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$i = (i^2)^2 = (-1)^2 = 1.$	**1**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify $\sqrt{(-16)}$. Student's answer: 4

Factor out i.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$\sqrt{(-16)} = 4i$. The student left out the imaginary unit.	**4i**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is i a real number? Student says: Yes

Think about what real numbers are.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
i is not on the real number line. It is an imaginary number, so it is not real.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add $(2 + 3i) + (4 - i)$. Student's answer: $6 + 4i$

Combine the real parts and the imaginary parts.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The imaginary parts are 3i and −i, which add to 2i. The answer is 6 + 2i.	**6 + 2i**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply $(3 + i)(3 - i)$. Student's answer: 8

Use the difference of squares.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$(3 + i)(3 - i) = 9 - i^2 = 9 + 1 = 10$. The student forgot that $i^2 = -1$.	**10**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply $(1 + 2i)(3 + i)$. Student's answer: $5 + 7i$

Use FOIL and replace i^2 with -1.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$2i \cdot i = 2i^2 = -2$. The answer is $3 + 7i - 2 = 1 + 7i$.	**$1 + 7i$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Subtract $(4 - 2i) - (1 + 5i)$. Student's answer: $3 + 3i$

Distribute the negative sign.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$-(1 + 5i) = -1 - 5i$. Then $-2i - 5i = -7i$, so the answer is $3 - 7i$.	**$3 - 7i$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify $(2i)^2$. Student's answer: 4

Square both the 2 and the i.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$(2i)^2 = 4i^2 = 4(-1) = -4.$	**-4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify $(2 + i)/(1 - i)$. Student's answer: $1/2 - 3/2\ i$

Multiply the top and bottom by the conjugate 1 + i.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$(2 + i)(1 + i) = 2 + 3i + i^2 = 1 + 3i$, and dividing by 2 gives $1/2 + 3/2\ i$. The student made a sign error on the imaginary part.	**$1/2 + 3/2\ i$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 + 4 = 0$. Student's answer: $x = \pm 2$

Solve for x^2 first, then take the square root.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$x^2 = -4$, so $x = \pm 2i$. The student dropped the imaginary unit.	**$x = \pm 2i$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 + 2x + 5 = 0$. Student's answer: $x = -1 \pm 2$

Use the quadratic formula.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The discriminant is 4 - 20 = -16, so the roots are (-2 ± 4i)/2 = -1 ± 2i.	**$x = -1 \pm 2i$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 - 6x + 10 = 0$. Student's answer: $x = 3 \pm 2i$

Use the quadratic formula.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The discriminant is 36 - 40 = -4, so $\sqrt{(-4)} = 2i$. Then $x = (6 \pm 2i)/2 = 3 \pm i$.	**$x = 3 \pm i$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 + 9 = 0$. Student's answer: $x = 9i$

Solve for x^2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$x^2 = -9$, so $x = \pm 3i$. The student squared the answer instead of taking the square root.	**$x = \pm 3i$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 + x + 1 = 0$. Student's answer: $x = (1 \pm i\sqrt{3})/2$

Use the quadratic formula carefully with the sign of b.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Here b = 1, so the formula gives $(-1 \pm i\sqrt{3})/2$. The student changed the sign of b.	**$x = (-1 \pm i\sqrt{3})/2$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 - 4x + 13 = 0$. Student's answer: $x = 2 \pm 6i$

Find the discriminant first.

What did the student do wrong? Why do you think the student made this error?

The discriminant is 16 − 52 = −36, so $\sqrt{(-36)} = 6i$. Then $x = (4 \pm 6i)/2 = 2 \pm 3i$.

Solve the problem correctly. Show all of your work.

$x = 2 \pm 3i$

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Rewrite $x^2 + 6x + 1$ by completing the square. Student writes: $(x + 3)^2 - 9$

Complete the square, then adjust the constant.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$(x + 3)^2 = x^2 + 6x + 9$, so $x^2 + 6x + 1 = (x + 3)^2 - 8$. The student forgot to adjust the +1.	**$(x + 3)^2 - 8$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the maximum value of $f(x) = -2x^2 + 8x$? Student's answer: It has a minimum of 8.

Decide whether the parabola opens up or down.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The coefficient -2 is negative, so the parabola opens down and has a maximum. The vertex is at x = 2 with f(2) = 8.	**Maximum of 8**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the vertex of $y = x^2 - 4x + 7$? Student's answer: (−2, 3)

Use x = -b/(2a).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
x = -(-4)/2 = 2. Then y = 4 - 8 + 7 = 3. The vertex is (2, 3).	**(2, 3)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Rewrite $2x^2 + 8x + 3$ in vertex form. Student writes: $2(x + 2)^2 + 5$

Factor 2 out of the first two terms, then complete the square.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$2(x + 2)^2 = 2x^2 + 8x + 8$, so the expression is $2(x + 2)^2 - 5$.	**$2(x + 2)^2 - 5$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The function $f(x) = (x - 3)^2 + 2$ is graphed. What is its minimum value? Student's answer: 3

Look at the constant added at the end.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The squared term is at least 0, so the smallest value is 0 + 2 = 2.	**2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Factor $2x^2 + 5x - 3$. Student writes: $(2x + 1)(x - 3)$

Check by multiplying.

What did the student do wrong? Why do you think the student made this error?

$(2x + 1)(x - 3) = 2x^2 - 5x - 3$. The correct factorization is $(2x - 1)(x + 3)$.

Solve the problem correctly. Show all of your work.

$(2x - 1)(x + 3)$

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Let $p(x) = x^2 + 3x - 4$. What is $p(1)$? Student's answer: 4

Substitute 1 for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$p(1) = 1 + 3 - 4 = 0$. The student did not finish the arithmetic.	**0**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the remainder when $x^3 - 2x + 1$ is divided by $x - 2$? Student's answer: 1

Use the remainder theorem: find p(2).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
p(2) = 8 - 4 + 1 = 5. The remainder is 5.	**5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is $x - 3$ a factor of $x^2 - 5x + 6$? Student says: No

Use the remainder theorem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$p(3) = 9 - 15 + 6 = 0$, so $x - 3$ is a factor.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is $x + 1$ a factor of $x^3 + 1$? Student says: No

Find p(-1).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
p(-1) = -1 + 1 = 0, so x + 1 is a factor.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the remainder when $x^2 + x$ is divided by $x - 1$? Student's answer: 0

Use the remainder theorem: find p(1).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
p(1) = 1 + 1 = 2. The remainder is 2.	**2**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Let $p(x) = 2x^3 - x + 5$. What is $p(-2)$? Student's answer: 9

Substitute -2 carefully.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$2(-8) - (-2) + 5 = -16 + 2 + 5 = -9$. The student dropped the negative sign.	**-9**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $6/x = 3$. Student's answer: $x = 3$

Multiply both sides by x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$6 = 3x$, so $x = 2$. The student divided 6 by 2 incorrectly.	**$x = 2$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x/(x - 2) = 2$. Student's answer: $x = 2$

Multiply both sides by x – 2, then check each answer.

What did the student do wrong? Why do you think the student made this error?

$x = 2(x - 2)$ gives $x = 4$. The value $x = 2$ makes the denominator zero, so it is not allowed.

Solve the problem correctly. Show all of your work.

$x = 4$

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $1/x + 1/2 = 1$. Student's answer: $x = 1$

Subtract 1/2, then take the reciprocal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
1/x = 1/2 gives x = 2. The student combined the fractions incorrectly.	**x = 2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $(x + 1)/3 = 4/x$. Student's answer: $x = 3$

Cross multiply and solve the quadratic.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$x(x + 1) = 12$ gives $x^2 + x - 12 = 0$, so $(x + 4)(x - 3) = 0$. Both $x = 3$ and $x = -4$ are solutions.	**$x = 3$ or $x = -4$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $3/(x - 1) = 6$. Student's answer: $x = 2.5$

Multiply both sides by x – 1.

What did the student do wrong? Why do you think the student made this error?

3 = 6(x – 1), so x – 1 = 1/2 and x = 1.5. The student added 1 instead of subtracting it.

Solve the problem correctly. Show all of your work.

x = 1.5

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x/2 + 3/x = 5/2$. Student's answer: $x = 2$ only

Multiply every term by 2x and solve the quadratic.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Multiplying gives $x^2 + 6 = 5x$, so $x^2 - 5x + 6 = 0$, which factors as $(x - 2)(x - 3) = 0$. Both $x = 2$ and $x = 3$ are solutions.	**$x = 2$ or $x = 3$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $y = x^2$ and $y = x + 2$. Student's answer: (2, 4) only

Set the two expressions equal and solve the quadratic.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$x^2 - x - 2 = 0$ gives $x = 2$ or $x = -1$. The second point is (−1, 1).	**(2, 4) and (−1, 1)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $y = x^2$ and $y = 4$. Student's answer: (2, 4)

Set $x^2 = 4$.

What did the student do wrong? Why do you think the student made this error?

$x^2 = 4$ gives $x = 2$ or $x = -2$. Both points (2, 4) and (−2, 4) work.

Solve the problem correctly. Show all of your work.

(2, 4) and (−2, 4)

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $y = x^2 - 1$ and $y = 3$. Student's answer: (2, 3)

Set $x^2 - 1 = 3$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$x^2 = 4$, so $x = \pm 2$. The solutions are (2, 3) and (−2, 3). The student left out one point.	**(2, 3) and (−2, 3)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $y = x^2$ and $y = 2x - 1$. Student's answer: (1, 2)

Set $x^2 = 2x - 1$ and solve.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$x^2 - 2x + 1 = 0$ gives $x = 1$. Then $y = 1^2 = 1$, so the point is (1, 1).	**(1, 1)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $y = x^2$ and $y = -x$. Student's answer: (0, 0) only

Set $x^2 = -x$ and factor.

What did the student do wrong? Why do you think the student made this error?

$x^2 + x = 0$ gives $x = 0$ or $x = -1$. The point (-1, 1) is also a solution.

Solve the problem correctly. Show all of your work.

(0, 0) and (-1, 1)

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many solutions does the system $y = x^2 + 1$ and $y = 0$ have? Student's answer: 2

Set $x^2 + 1 = 0$ and check for real solutions.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$x^2 = -1$ has no real solutions, so the graphs do not intersect. The answer is 0.	**0**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the x-values where $y = x^2$ and $y = 2x + 3$ intersect. Student's answer: $x = 3$

Set the expressions equal and solve.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$x^2 - 2x - 3 = 0$ gives $x = 3$ or $x = -1$. The student missed $x = -1$.	**$x = 3$ or $x = -1$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find where y = 2^x and y = x + 3 intersect. Student says: They intersect at x = 3.

Compare values at integer x-values.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
At x = 3, $2^3 = 8$ but 3 + 3 = 6. The graphs cross between x = 2 and x = 3, not at x = 3.	**Between x = 2 and x = 3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the x-values where $y = x^3$ and $y = x$ intersect. Student's answer: $x = 1$ only

Set $x^3 = x$ and factor.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$x^3 - x = 0$ gives $x(x - 1)(x + 1) = 0$, so $x = 0, 1,$ or -1.	**$x = -1, 0, 1$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the intersection of $y = \sqrt{x}$ and $y = x/2$. Student's answer: (4, 2)

Square both sides and solve.

What did the student do wrong? Why do you think the student made this error?

$\sqrt{x} = x/2$ gives $x = x^2/4$, so $x = 0$ or $x = 4$.
The point (0, 0) is also an intersection.

Solve the problem correctly. Show all of your work.

(0, 0) and (4, 2)

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the intersection of $y = 1/x$ and $y = x$ for positive x. Student's answer: $x = 1$

Set $1/x = x$ and solve.

What did the student do wrong? Why do you think the student made this error?

$x^2 = 1$ gives $x = 1$ or $x = -1$. For positive x, the intersection is at $x = 1$. The student's answer is correct for positive x, but the task asked for any real x, which adds $x = -1$.

Solve the problem correctly. Show all of your work.

$x = 1$ or $x = -1$

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the x-value where y = 3^x and y = 9 intersect. Student's answer: x = 3

Write 9 as a power of 3.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$9 = 3^2$, so x = 2. The student used 3 instead of 2.	**x = 2**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the average rate of change of $f(x) = x^2$ on $[1, 3]$. Student's answer: 8

Use (f(b) − f(a)) ÷ (b − a).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
f(3) − f(1) = 9 − 1 = 8. Divide by 3 − 1 = 2 to get 4.	**4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the average rate of change of $f(x) = 3x - 2$ on $[0, 4]$. Student's answer: 12

Divide the change in f by the change in x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided the change in f by 1 instead of 4. $f(4) - f(0) = 12$, and $12 \div 4 = 3$.	**3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the average rate of change of $f(x) = 2$ on $[0, 2]$. Student's answer: 3

Use (f(2) – f(0)) ÷ 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not divide by 2. f(2) – f(0) = 4 – 1 = 3, and 3 ÷ 2 = 1.5.	**1.5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

table shows f(1) = 5 and f(3) = 11. What is the average rate of change from x = 1 to x = 3? Student's answer:

Divide the change in output by the change in input.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The change in output is 6, and the change in input is 2. 6 ÷ 2 = 3.	**3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the average rate of change of $g(x) = x^2 - 4x$ on $[0, 5]$. Student's answer: 5

Find g(5) and g(0), then divide by 5.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
g(5) = 25 - 20 = 5, and g(0) = 0. The change is 5, and 5 ÷ 5 = 1.	**1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the average rate of change of $f(x) = 1/x$ on $[1, 2]$. Student's answer: 1/2

Use (f(2) − f(1)) ÷ (2 − 1).

What did the student do wrong? Why do you think the student made this error?

f(2) − f(1) = 1/2 − 1 = −1/2. The rate is negative because the function decreases.

Solve the problem correctly. Show all of your work.

−1/2

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Where is the vertex of $f(x) = |x - 2| + 1$? Student's answer: $(-2, 1)$

Find the value that makes the absolute value zero.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$|x - 2| = 0$ when $x = 2$, so the vertex is $(2, 1)$. The student changed the sign of 2.	**$(2, 1)$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the domain of $f(x) = \sqrt{(x + 4)}$? Student's answer: $x \geq 4$

Make the expression under the square root nonnegative.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$x + 4 \geq 0$ gives $x \geq -4$. The student changed the sign.	**$x \geq -4$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Let $f(x) = x + 1$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. What is $f(-2)$? Student's answer: 4

Choose the correct piece for x = −2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Since −2 < 0, use the first piece. f(−2) = −2 + 1 = −1.	**−1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Describe the graph of $y = -2|x|$. Student says: It opens upward with vertex (0, 0).

Look at the sign in front of the absolute value.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The negative coefficient reflects the graph over the x-axis, so it opens downward.	**Opens downward with vertex (0, 0)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the maximum value of $f(x) = -|x| + 3$? Student's answer: -3

Find where the absolute value is zero.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$|x|$ is 0 at $x = 0$, so $f(0) = 3$. The maximum is 3.	**3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is $f(-2)$ if $f(x) = x^3$? Student's answer: 8

Cube the input.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$(-2)^3 = -8$. The sign is kept when cubing a negative number.	**-8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 8 equivalent to 2^(3x)? Student says: No

Write 8 as a power of 2.

What did the student do wrong? Why do you think the student made this error?

$8 = 2^3$, so $8 = (2^3) = 2^{(3x)}$. The two expressions are equivalent.

Solve the problem correctly. Show all of your work.

Yes

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A population is modeled by y = 1000(1.05)^t. What is the annual growth rate? Student's answer: 1.05%

Subtract 1 from the base and write as a percent.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The base is 1.05, so the growth rate is 0.05, or 5% per year. The student used the base as a percent.	**5%**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Rewrite 3^(2t) in the form b^t. Student's answer: 6^t

Use the power rule (a^m)^n = a^(mn).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
3^(2t) = (3²)^t = 9^t. The student multiplied the bases.	**9^t**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A value is modeled by y = 200(0.75)^t. What is the annual rate of decay? Student's answer: 75%

Subtract the base from 1.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The decay rate is 1 - 0.75 = 0.25, or 25% per year. 75% is the amount remaining.	**25%**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Rewrite 4 in the form 2^(kx). Student's answer: 2^(x/2)

Write 4 as a power of 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$4 = 2^2$, so $4 = 2^{(2x)}$. The student divided the exponent instead of multiplying.	**$2^{(2x)}$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

mple is modeled by y = 80(1/2)^(t/5), where t is in years. How much remains after 10 years? Student's answ

Substitute t = 10 and simplify the exponent.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
10/5 = 2, so y = 80(1/2)² = 80/4 = 20. The student used only one half.	**20**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

x + 5 and g has a constant rate of change of 3. Which function has the greater rate of change? Student's a

Compare the slopes or rates.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
f has rate 2 and g has rate 3. Function g has the greater rate.	**g**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which grows faster eventually: $f(x) = 2$ or $g(x) = x^2$? Student's answer: $g(x) = x^2$

Compare large values of x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
At $x = 10$, $2^1 = 1024$ while $10^2 = 100$. Exponential functions eventually grow faster.	**$f(x) = 2$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$y = 4x + 1$ and $y = x^2 + 3$ are compared. Which has the greater y-intercept? Student's answer: $y = 4x + 1$

Find y when x = 0 for each function.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The linear function has y-intercept 1. The quadratic has y-value 0 + 3 = 3 when x = 0, so $y = x^2 + 3$ has the greater intercept.	**$y = x^2 + 3$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which eventually grows faster: $f(x) = 3x$ or $g(x) = 0.1 \cdot 2$? Student's answer: $f(x) = 3x$

Compare growth for large x.

What did the student do wrong? Why do you think the student made this error?

Exponential growth eventually overtakes linear growth. g(x) grows faster for large x.

Solve the problem correctly. Show all of your work.

$g(x) = 0.1 \cdot 2$

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

) = 2x + 1 has a constant rate. $g(x) = x^2$ does not. Which has a constant rate of change? Student's answer: g

Think about which graph is a straight line.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Linear functions have a constant rate of change. f(x) has rate 2 everywhere.	**f(x)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

aximum value 6, and g has a graph with maximum value 8. Which has the greater maximum? Student's a

Compare the maximum values.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
8 is greater than 6, so g has the greater maximum.	**g**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Let $f(x) = x^2$ and $g(x) = f(x) + 3$. What is $g(2)$? Student's answer: 1

Find f(2), then add 3.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 3. f(2) = 4, so g(2) = 4 + 3 = 7. The graph shifts up 3.	**7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Let $g(x) = f(x - 2)$ with $f(x) = x^2$. What is $g(3)$? Student's answer: 9

Substitute 3 - 2 into f.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not subtract 2. g(3) = f(1) = 1.	**1**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Let $g(x) = 2f(x)$ with $f(x) = x^2$. What is $g(3)$? Student's answer: 9

Find f(3), then multiply by 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not multiply by 2. f(3) = 9, so g(3) = 18.	**18**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Let $g(x) = -f(x)$ with $f(x) = x^2$. What is $g(2)$? Student's answer: 4

Find f(2), then take the opposite.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student lost the negative sign. g(2) = -4. The graph reflects over the x-axis.	**-4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Let $g(x) = f(x + 1)$ with $f(x) = |x|$. What is $g(-3)$? Student's answer: -4

Substitute −3 + 1 into f.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$g(-3) = f(-2) = |-2| = 2$. The student subtracted incorrectly.	**2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the vertex of $g(x) = (x - 1)^2 - 4$? Student's answer: (-1, -4)

Compare with vertex form $(x - h)^2 + k$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Here h = 1 and k = -4, so the vertex is (1, -4). The student changed the sign of h.	**(1, -4)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the inverse of $f(x) = 2x + 3$. Student writes: $f^{1}(x) = (x + 3)/2$

Swap x and y, then solve for y.

What did the student do wrong? Why do you think the student made this error?

y = 2x + 3 gives x = 2y + 3, so y = (x - 3)/2. The student added 3 instead of subtracting.

Solve the problem correctly. Show all of your work.

$f^{1}(x) = (x - 3)/2$

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $f(x) = x^3$, what is $f^1(8)$? Student's answer: 4

Find the x that gives f(x) = 8.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student squared 8 instead of taking a cube root. Since $2^3 = 8$, $f^1(8) = 2$.	**2**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is $f(x) = x^2$ one-to-one over all real numbers? Student says: Yes

Check whether two inputs can give the same output.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
f(2) = f(-2) = 4. Two inputs give one output, so f is not one-to-one and has no inverse on all reals.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $f(x) = 3x - 6$, what is $f'(0)$? Student's answer: -2

Set the output to 0 and solve for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
3x - 6 = 0 gives x = 2. The student flipped the sign.	**2**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the inverse of $f(x) = (x + 1)/2$. Student writes: $f^{1}(x) = 2x + 1$

Solve $y = (x + 1)/2$ for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$x = 2y - 1$, so $f^{1}(x) = 2x - 1$. The student forgot to subtract 1.	**$f^{1}(x) = 2x - 1$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $f(x) = 2x$, what is $f^{1}(f(5))$? Student's answer: 10

Apply f^{1} to the output of f.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A function and its inverse undo each other. $f^{1}(f(5)) = 5$.	**5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is log 8? Student's answer: 4

Ask what power of 2 gives 8.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$2^3 = 8$, so log 8 = 3. The student used $2^2 \times 2$.	**3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is log 1000? Student's answer: 2

Count the powers of 10.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$10^3 = 1000$, so the answer is 3. The student counted only two zeros.	**3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve 2 = 32. Student's answer: x = 4

Write 32 as a power of 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
2 = 32, so x = 5. The student used 2 = 16 as the target.	**x = 5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve 3 = 81. Student's answer: x = 3

Write 81 as a power of 3.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
3 = 81, so x = 4. The student confused 81 with 27.	**x = 4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is log 1? Student's answer: 1

Ask what power of 5 gives 1.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Any base to the power 0 is 1, so log 1 = 0.	**0**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate log 27 - log 3. Student's answer: 3

Use the quotient rule or evaluate each term.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
log 27 = 3 and log 3 = 1, so the difference is 2. The student did not subtract the second term.	**2**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Convert 180° to radians. Student's answer: 2π

Use 180° = π radians.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
180° equals π radians. The student used the value for a full circle, which is 360°.	**π**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Convert 90° to radians. Student's answer: π/3

Divide 90 by 180 and multiply by π.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
90/180 = 1/2, so 90° = π/2 radians.	**π/2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

An arc has radius 5 and central angle 2 radians. What is the arc length? Student's answer: 7

Use s = r.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added r and . The arc length is 5 × 2 = 10.	**10**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Convert π/6 radians to degrees. Student's answer: 60°

Multiply by 180/π.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
(π/6)(180/π) = 30. The student doubled the angle.	**30°**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Convert 45° to radians. Student's answer: π/2

Use 45/180 times π.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
45/180 = 1/4, so 45° = π/4 radians.	**π/4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

An arc has radius 3 and central angle π/3 radians. What is the arc length? Student's answer: π/3

Use s = r.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave the angle. The arc length is 3 × π/3 = π.	**π**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is $\cos(\pi/2)$? Student's answer: 1

Think about the point on the unit circle at π/2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
At π/2 (90°), the point on the unit circle is (0, 1). The cosine is the x-coordinate, which is 0.	**0**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is sin π? Student's answer: 1

Think about the point on the unit circle at π.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
At π (180°), the point is (-1, 0). The sine is the y-coordinate, which is 0.	**0**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the point on the unit circle at 60º? Student's answer: $(\sqrt{3}/2, 1/2)$

Recall the 30-60-90 triangle.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
At 60°, cos = 1/2 and sin = √3/2. The point is (1/2, √3/2).	**(1/2, √3/2)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is sin(3π/2)? Student's answer: 1

Find the point at 3π/2 on the unit circle.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
At 3π/2 (270°), the point is (0, -1). The sine is -1.	**-1**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is cos π? Student's answer: 0

Find the point at π on the unit circle.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
At π, the point is (-1, 0). The cosine is the x-coordinate, which is -1.	**-1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is tan(π/4)? Student's answer: √2

Use tan = sin/cos at π/4.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
At π/4, sin = cos = √2/2, so tan = 1. The student confused tan with sin.	**1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the amplitude of $y = 3\sin(2x)$? Student's answer: 2

Find the coefficient in front of the sine.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The amplitude is $\|3\| = 3$. The 2 is inside the sine and affects the period.	**3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the period of $y = \sin(2x)$? Student's answer: 2π

Use period = $2\pi/B$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The period is $2\pi/2 = \pi$. The student did not divide by B.	**π**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the period of $y = \cos(x/3)$? Student's answer: 3π

Use period = 2π/B.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Here B = 1/3, so the period is 2π ÷ (1/3) = 6π. The student divided by 3 instead of by 1/3.	**6π**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the midline of $y = 4 + 2 \cos x$? Student's answer: 2

Find the constant added outside the cosine.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The midline is y = 4, the constant added to the cosine. The 2 is the amplitude.	**y = 4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the maximum value of $y = 5 \sin x - 1$? Student's answer: 5

Add the vertical shift to the amplitude.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The sine reaches 1 at most, so the maximum is 5(1) - 1 = 4.	**4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A tide model has a maximum of 12 feet and a minimum of 4 feet. What is the amplitude? Student's answer:

Find half the distance between the maximum and minimum.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the full distance. The amplitude is (12 - 4) ÷ 2 = 4 feet.	**4 feet**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A model is $y = 2x + 1$. What is the predicted y-value when $x = 5$? Student's answer: 10

Substitute 5 into the model.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$2(5) + 1 = 11$. The student forgot to add 1.	**11**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

An actual value is 12 and the predicted value is 11. What is the residual? Student's answer: -1

Residual = actual - predicted.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
12 - 11 = 1. The student reversed the subtraction.	**1**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

plot has a correlation coefficient of r = 0.95. Is the linear relationship weak or strong? Student's answer: wea

Think about how close r is to 1.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
r is close to 1, which indicates a strong positive linear relationship.	**Strong positive**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A linear model has slope 3. What happens to y when x increases by 1? Student's answer: y decreases by 3

Interpret the sign of the slope.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A positive slope means y increases by 3 for each increase of 1 in x.	**y increases by 3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

a points curve upward and get steeper. Which model fits better: linear or exponential? Student's answer: lin

Look at the shape of the data.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Data that gets steeper as x increases fits an exponential model better than a line.	**Exponential**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A residual plot shows a clear curved pattern. Is a linear model appropriate? Student says: Yes

Think about what a residual pattern means.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A curved pattern in the residuals means the linear model is missing a trend, so a linear model is not appropriate.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

taxi cost is C = 25 + 4m, where m is minutes. What does the slope 4 mean? Student's answer: $25 per minu

Identify the rate of change.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The slope is the rate of change, which is $4 per minute. The 25 is the starting fee.	**$4 per minute**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

the same taxi model C = 25 + 4m, what does the intercept 25 mean? Student's answer: the cost per minut

Think about the value when m = 0.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
When m = 0, the cost is $25, which is the fixed starting fee. The cost per minute is the slope.	**The starting fee**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

odel is T = 80 - 2h, where h is hours. What does the slope mean? Student's answer: The temperature rises 2

Check the sign of the slope.

What did the student do wrong? Why do you think the student made this error?

The slope is -2, so the temperature falls 2 degrees per hour.

Solve the problem correctly. Show all of your work.

The temperature falls 2 degrees per hour

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

model is P = 500 + 30t, where t is years. What does the intercept mean? Student's answer: the growth rate

Think about the value when t = 0.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The intercept 500 is the starting population. The growth rate is the slope of 30.	**The starting population of 500**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A distance model is d = 12t, where t is hours. What is the unit of the slope? Student's answer: hours per mile

Think about distance divided by time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The slope is distance per hour, which is 12 miles per hour.	**Miles per hour**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

odel is S = 1500 - 50w, where w is weeks. What does the slope mean? Student's answer: Sales increase $50

Check the sign of the slope.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The slope is -50, so sales decrease by $50 per week.	**Sales decrease $50 per week**